Significant Figure Rules

CHEMISTRY REFERENCE

Which digits count

Rule	Example	Sig figs
Nonzero digits are always significant.	4.72	**3**
Zeros between nonzero digits are significant.	2005	**4**
Leading zeros are never significant. They only place the decimal point.	0.0032	**2**
Trailing zeros are significant when the number has a decimal point.	3.200 · 150.	**4 · 3**
Trailing zeros with no decimal point are not significant.	1500	**2**
In scientific notation, every digit in the coefficient is significant.	4.50×10^3	**3**
Exact numbers (counted items, defined conversions) have unlimited significant figures.	12 eggs · 1 m = 100 cm	**∞**

Calculations

Multiplying and dividing

The answer has the same number of **significant figures** as the measurement with the fewest.

$2.5\ \text{cm} \times 3.47\ \text{cm} = 8.675 \rightarrow \mathbf{8.7\ cm^2}$

2.5 has 2 significant figures, so the answer has 2.

Adding and subtracting

The answer has the same number of **decimal places** as the measurement with the fewest.

$12.11\ \text{g} + 3.2\ \text{g} = 15.31 \rightarrow \mathbf{15.3\ g}$

3.2 has 1 decimal place, so the answer has 1.

Rounding

Look at the first digit you are dropping. 5 or more: round up. Less than 5: leave the last kept digit alone. 3.476 to 3 sig figs → 3.48 · 3.474 → 3.47

In a multi-step problem, keep extra digits until the final answer, then round once.